

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (E&C) (Credit System) Degree Examinations
Supplementary Examination - May - June 2023

19EC701 – POWER ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit***

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) With neat diagram and relevant waveforms explain the switching characteristics of i) Power BJT ii) Power MOSFET	10	L*2	1	1
	b) The beta β of bipolar transistor varies from 12 to 75. The load resistance is $R_C = 1 \Omega$. The dc supply voltage is $V_{CC} = 30 \text{ V}$ and the input voltage to the base circuit is $V_B = 5 \text{ V}$. If $V_{CE(sat)} = 1.2 \text{ V}$, $V_{BE(sat)} = 1.6 \text{ V}$, and $R_B = 0.8 \Omega$, determine (i) the ODF, (ii) the forced β and (iii) the power loss in the transistor P_T , (iv) the value of R_B that will result in saturation with ODF of 5	10	L3	1	1
2.	a) With neat diagram and relevant waveforms explain i) Fast Gate Turn-on control ii) Proportional Base drive control	10	L2	1	1
	b) A Bipolar Junction Transistor is used as a switch. The parameters are $V_{CC} = 250 \text{ V}$, $V_{BE(sat)} = 3 \text{ V}$, $I_B = 8 \text{ A}$, $V_{CE(sat)} = 2 \text{ V}$, $I_{CS} = 100 \text{ A}$, $t_d = 0.5 \mu\text{s}$, $t_r = 1 \mu\text{s}$, $t_s = 5 \mu\text{s}$, $t_f = 3 \mu\text{s}$, and $f_s = 10 \text{ kHz}$. The duty cycle is $k = 50\%$. The collector-to-emitter leakage current is $I_{CEO} = 3 \text{ mA}$. Determine the power loss due to collector current (i) during turn-on $t_{on} = t_d + t_r$, (ii) during conduction period t_n ,	10	L3	1	1
		Unit – II			
3.	a) With neat circuit diagram and relevant waveforms explain RC-Triggering of Thyristor. Write the requirements for selecting Resistor and Capacitor for proper triggering control.	10	L2	2	1
	b) With relevant switching diagrams explain dynamic turn-on and turn-off characteristics of SCR.	10	L2	2	1
4.	a) With neat circuit diagram and relevant waveforms explain Class-B commutation of Thyristor. Write the expressions for Thyristor conduction time, Capacitor voltage at the end of conduction time and Circuit turn-off time.	10	L2	2	1
	b) With the help of two transistor analogy and internal capacitance diagram explain the various turn-on methods of the SCR.	10	L2	2	1
		Unit – III			
5.	a) With neat circuit diagram and relevant waveforms explain the operation of Single-Phase Dual Converter with circulating current. Write the expression for instantaneous circulating current.	10	L2	3	1
	b) A single-phase full-wave ac voltage controller with two anti-parallel thyristors has a resistive load of $R = 10 \Omega$ and the input voltage is $V_s = 120 \text{ V (rms)}$, 60 Hz . The delay angles of thyristors T1 and T2 are equal: $\alpha_1 = \alpha_2 = \alpha = \pi/2$. Determine (i) the rms output voltage V_o , (ii) the input PF, (iii) the average current of thyristors I_A , and (iv) the rms current of thyristors I_R .	10	L3	3	2

P.T.O.

6. a) With neat circuit diagram and relevant waveforms explain the operation of Single-Phase Full Converter. Write the expression for output dc voltage.
- b) A single-phase full-wave ac voltage controller with two anti-parallel thyristors has a resistive load of $R = 10 \Omega$ and the input voltage is $V_s = 120 \text{ V (rms)}$, 60 Hz . The thyristor switches are on for $n=25$ cycles and off for $m=75$ cycles. Determine (i) the rms output voltage V_o , (ii) the input PF, (iii) the average current of thyristors I_A , and (iv) the rms current of thyristors I_R .

10	L2	3	1
10	L3	3	2

Unit – IV

7. a) With neat circuit diagram and relevant waveforms explain the operation of Buck Regulator.
Derive the expression for Output voltage V_o in-terms of Input voltage V_s and duty cycle k . Assume CCM.
- b) Design a DC-DC Converter for the following specifications:
Input Voltage $V_s = 10\text{V}$, Switching Frequency $F_s=10\text{kHz}$, Average Output Voltage $V_a = -4\text{V}$, Average Load current $I_a=1.25\text{A}$, $\Delta V = 2\%$ and $\Delta I = 30\%$.
Also Find the critical values of L & C and the peak current through the transistor.
8. a) With neat circuit diagram and relevant waveforms explain the operation of Boost Regulator.
Derive the expression for Output voltage V_o in-terms of Input voltage V_s and duty cycle k . Assume CCM.
- b) Design a DC-DC Converter for the following specifications:
Input Voltage $V_s = 10\text{V}$, Switching Frequency $F_s=10\text{kHz}$, Average Output Voltage $V_a = -15\text{V}$, Average Load current $I_a=1.25\text{A}$, $\Delta V = 2\%$ and $\Delta I = 30\%$.
Also find the critical values of L & C and the peak current through the transistor.

10	L2	4	1
10	L3	4	2
10	L2	4	1
10	L3	4	2

Unit – V

9. a) With neat circuit diagram and relevant waveforms explain the operation of Single-Phase Bridge Inverter.
Write the expressions for output rms voltage and Fourier series expression for Instantaneous output voltage.
- b) Explain the concept of
- Phase-Displacement control
 - Sinusoidal Pulse-Width Modulation
10. a) With neat circuit diagram and relevant waveforms explain the operation of Current Source Inverter.
Write the expressions for output rms voltage and Fourier series expression for Instantaneous output voltage.
- b) Explain the concept of
- Modified Sinusoidal Pulse-Width Modulation
 - Multiple Pulse-Width Modulation

10	L2	5	1
10	L2	5	1
10	L2	5	1
10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
